

A Project Report
on
Diabetes Prediction
using Machine Learning

"A Comparative Study of Logistic Regression and Random Forest"

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

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1. Executive Summary

This project focuses on predicting whether a person is diabetic or not using machine learning techniques. The problem is treated as a **binary classification task**. Two models - **Logistic Regression** and **Random Forest Classifier** - were implemented and compared. After preprocessing and training, **Logistic Regression performed better**, achieving higher accuracy and better prediction capability. The project demonstrates how preprocessing, model selection, and tuning significantly impact performance.

2. Problem Understanding

2.1. Problem Description

This project aims to predict diabetes in patients based on medical attributes. It is a **classification problem** where the target variable is binary (0 = Non-Diabetic, 1 = Diabetic). Input features include glucose level, BMI, insulin, blood pressure, etc. The goal is to build a model that accurately classifies patients and helps in early detection of diabetes.

2.2. SDGs

- **SDG 3: Good Health and Well-being**
Helps in early diagnosis and prevention of diabetes using data-driven methods.

2.3. Dataset

2.3.1. Dataset: **Diabetes Dataset (PIMA Indians Diabetes Dataset)**

2.3.2. Features include:

- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Age, etc.

2.3.3. Target variable: **Outcome (0 or 1)**

2.4. Algorithms Used

- Logistic Regression
- Random Forest Classifier

2.5. Performance Metrics

- Accuracy Score
- Confusion Matrix
- Classification Report (Precision, Recall, F1-score)
- ROC Curve & AUC

3. Data Handling & Preprocessing

- Loaded dataset using Pandas
- Identified zero values in important columns (Glucose, BMI, etc.)
- Replaced invalid zero values with **NaN**
- Filled missing values using **median imputation**
- Split dataset into training (80%) and testing (20%)
- Applied **Standard Scaling** to normalize data

4. Model Development & Training

4.1. Logistic Regression

- Trained using scaled data
- Works well for linear relationships

4.2. Random Forest

- Ensemble model using multiple decision trees
- Captures complex patterns and non-linear relationships

Both models were trained and predictions were made on test data.

5. Hyperparameter Tuning

→ Used **GridSearchCV** for tuning both models

5.1. Logistic Regression:

- Tuned parameter: C (regularization strength)

5.2. Random Forest:

5.2.1 Tuned parameters:

- Number of trees (n_estimators)
- Maximum depth (max_depth)

5.2.2. Created comparison tables to track performance across trials

5.2.3. Best parameters selected based on highest accuracy

6. Evaluation & Interpretation

6.1. Compared both models using accuracy:

- Logistic Regression → Better performance
- Random Forest → Moderate performance

6.2. Confusion Matrix showed that Logistic Regression achieved better classification results with fewer misclassifications

6.3. ROC Curve indicated strong predictive capability

6.4. Feature Importance revealed key factors:

- Glucose
- BMI
- Age

Insight: Logistic Regression performed better because the relationship between the features (Glucose, BMI, Age) and the diabetes outcome is largely linear, and with only 768 samples, the simpler model generalises better without overfitting

7. Conclusion & Learning Outcomes

- Logistic Regression is more suitable for this dataset
- Model shows good prediction capability and can be used as a base for real-world healthcare applications
- With further tuning and more data, it can become production-ready

Reflection

This project helped bridge the gap between theory and practical implementation by:

- Applying preprocessing techniques on real data
- Understanding model differences
- Learning hyperparameter tuning

- Interpreting model results using visualizations

It strengthened concepts of **classification, data cleaning, and model evaluation** in a real-world context.